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SUPERSEDING
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MILITARY SPECIFICATION

FLASHERS, AIRCRAFT NAVIGATIONAL LIGHT GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for solid state aircraft flashers, which are used for turning aircraft position lights on and off periodically. Flashers shall have the load current controlled with a transistor, controlled rectifier, magnetic amplifier or equivalent device. Flashers shall have either single or dual output circuits.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallet, Materials-handling, Wood (General Construction Requirements).
QQ-S-781	-	Strapping, Steel, Flat and Seals.
QQ-Z-325	-	Zinc Coating, Electrodeposited, Requirements for.
PPP-B-566	-	Boxes, Folding Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape, Packaging, Waterproof.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-P-116	-	Preservation-Packaging, Method of.
MIL-G-5572	-	Gasoline, Aviation: Grades 80/87, 100/130, 115/145.
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance.
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-A-8625	-	Anodic Coatings, For Aluminum and Aluminum Alloys.
MIL-B-43014	-	Boxes, Water Resistant; Paperboard, Folding, Set-up and Metal-Stayed.
MIL-C-45662	-	Calibration Systems Requirements.

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads on 40" x 48" Pallets.
MIL-STD-454 - Standard General Requirement for Electronic Equipment.
MIL-STD-461 - Electromagnetic Interference, Characteristics Requirements for Equipment.
MIL-STD-704 - Electric Power, Aircraft, Characteristics and Utilization of.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Military standards. The individual item requirements shall be as specified herein and in accordance with the applicable military standards and specification sheet. In the event of any conflict between requirements of this specification and the military standards or specification sheet the latter shall govern.

3.2 Qualification. Flashers furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.2).

3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the flashers to meet the performance requirements of this specification. Materials used shall be self-extinguishing, and shall not support combustion, give off noxious gases in harmful quantities, give off gases in quantities sufficient to cause explosion of sealed enclosures, or form current-carrying tracks when subjected to any of the tests specified herein. Unless otherwise specified (see 3.1), the selection of material shall be such as to provide a shelf life of at least 2 years without affecting the operation of the flasher. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion.

3.3.1.1 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metal in contact, which tends toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable. However, metal spraying or metal plating of dissimilar base metals to provide similar to suitable abutting surfaces is permitted. Dissimilar metals shall be as defined in 6.5 through 6.5.4, inclusive, and table V. In hermetic seals, the 0.25 volt difference between the header material and the housing material is not applicable.

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3.3.1.2 Aluminum alloy parts. Aluminum alloy parts, where practicable, shall be anodically treated in accordance with MIL-A-8625. The aluminum oxide film deposited by this treatment shall be removed from actual contact area of all surfaces required to act as a path for electrical power and from local areas under screws, nuts, etc., used by assembly or mounting purposes to provide an adequate bonding connection.

3.3.2 Fungus-resistant. Materials used in construction of flashers shall be fungus inert (see requirement 4 of MIL-STD-454).

3.3.3 Plating. Metals shall be of a corrosion-resistant type, or shall be plated or treated to resist corrosion. Cadmium plating shall not be used for plating any part of the flasher while zinc and unfused tin plating shall not be used for plating internal parts of the flasher. Zinc plating shall be in accordance with class I, type II of QQ-Z-325.

3.4 Design and construction. Flashers shall be of the design, construction, weight and physical dimensions specified (see 3.1). Flashers shall be designed to insure proper operation when mounted in any position. Construction shall be such as to take full advantage of the design for minimum size and weight, long life, maximum reliability and minimum radio noise.

3.4.1 Envelope. When specified (see 3.1), flashers shall be provided with a means to permit assured contact to case ground through a secure low resistance connection. The enclosure shall not be electrically connected to any terminal unless specified.

3.4.2 Paralleled semiconductors. Flashers shall not utilize paralleled semiconductors for carrying rated lamp load.

3.4.3 Dimensions. The overall dimensions of the flasher shall be as specified (see 3.1).

3.4.4 Weight. The weight of the flasher shall be held to the minimum practicable but shall not exceed the specified maximum weight (see 3.1).

3.4.5 Mounting means. The mounting means shall be as specified (see 3.1).

3.4.6 Threaded parts. All threaded parts shall be in accordance with Handbook H28. Where practical, all threads shall be in conformity with the coarse-thread series. The fine-thread series may be used only for applications that show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch between 16 and 36, which is used in the fine thread series. Terminal threads shall be class 2A and 2B for external and internal threads, respectively.

3.4.7 Functional diagram. A connection diagram of the external wiring of the flasher shall be provided. The diagram shall identify each terminal, shall be permanently attached to the flasher, shall remain legible and securely attached to the flasher during all tests required by this specification.

3.4.8 Fail safe. Flashers shall be so designed that with rated voltage applied to the input terminals, only internal failure of the flasher timing circuitry will result in continuous rated power output from the output terminals.

3.4.9 Encapsulation. Unless otherwise specified (see 3.1), flashers shall be encapsulated.

3.5 Electrical requirements.

3.5.1 Input power. The flashers shall meet all applicable requirements of this specification when operating from power sources as defined in specification MIL-STD-704, category B. Voltage transients having magnitudes and time durations as defined therein shall not damage the flashers.

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3.5.2 Input voltage. The rated input voltage of each type of flasher shall be as indicated (see 3.1). When the rated input voltage is given as 28 volts ac or dc, the limiting input voltage for performance requirements shall be 26.3 to 29 volts. When the rated input voltage is given as 115 volts, 400 hertz (Hz), the limiting input voltage and frequency for performance requirement shall be 107.5 to 119.5 volts, RMS, and 380 to 420 Hz, respectively.

3.5.3 Load requirement. Unless otherwise specified (see 3.1), flashers shall be capable of operating the applicable incandescent lamp load specified in table I. Unless otherwise specified, lamp ratings are considered steady state current values.

TABLE I. Lamp rating.

Source voltage	Load range (amperes)
28V dc	1.0 to 7.5
28V 400 Hz ac	1.0 to 5.0
115V 400 Hz ac	0.25 to 4.0
28V ac or dc	1.0 to 15.0

3.5.4 Flash characteristics:

3.5.4.1 Flash rate. The flash rate shall be as specified in 3.5.4.1.1 and 3.5.4.1.2, as applicable.

3.5.4.1.1 Single output circuits. When flashers are tested as specified in 4.5.2.1.1, the flash rate shall be as specified (see 3.1).

3.5.4.1.2 Dual output circuits. When flashers are tested as specified in 4.5.2.1.2, the flash rate shall be as specified (see 3.1).

3.5.4.2 Flash cycle (on to off ratio). The "ON" time of the flash cycle shall be considered that portion of the cycle during which the voltage supplied to the lamp load is 90 percent or more of the rated voltage. The "OFF" time shall be considered that portion when the voltage supplied to the load is less than 10 percent of rated voltage. The flash cycle shall be as specified in 3.5.4.2.1 and 3.5.4.2.2, as applicable.

3.5.4.2.1 Single output circuits. When flashers are tested as specified in 4.5.2.2.1, the "ON" time shall be 70 ±10 percent of the total cycle.

3.5.4.2.2 Dual output circuits. When flashers are tested as specified in 4.5.2.2.2, the "ON" time of one circuit shall equal the "OFF" time of the other and vice versa within ±5 percent.

3.5.4.3 Switching time. When flashers are tested as specified in 4.5.2.3, the switching time shall be as specified in 3.5.4.3.1 and 3.5.4.3.2, as applicable.

3.5.4.3.1 DC flashers. The elapsed time for switching the "OFF" to "ON" or "ON" to "OFF" condition shall not be greater than 0.03 second for dc flashers.

3.5.4.3.2 AC flashers. Zero crossover switching shall be used for initiation and termination of the "ON" time of each cycle. Initiation of the "ON" time shall coincide with zero voltage crossover and termination of "ON" time shall coincide with zero current crossover within 20 degrees, respectively.

3.6 Voltage drop. Flashers shall be tested as specified in 4.5.3 and unless otherwise specified (see 3.1), the voltage drop within the flasher shall not exceed 4.0 percent of the input voltage for any value of load within the specified limits.

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3.7 High and low temperature soak and operation. When tested as specified in 4.5.4, flashers shall operate satisfactorily when tested under the following conditions.

3.7.1 High temperature. At 71°C (unless otherwise specified) (see 3.1), following the soak period specified in 4.5.4.1, flashers shall meet the requirements of 3.5.4.1, 3.5.4.2, 3.5.4.3, and 3.6, as applicable.

3.7.2 Low temperature. At -62°C and -54°C, following the soak period specified in 4.5.4.2, flashers shall meet the requirements of 3.5.4.1, 3.5.4.2, 3.5.4.3, and 3.6, as applicable. There shall be no distortions, cracking, differential expansion or pliability of nonmetallic parts.

3.8 Vibration. After flashers are tested as specified in 4.5.5, they shall meet the requirements of 3.5.4.1, 3.5.4.2, 3.5.4.3, and 3.6, as applicable.

3.9 Immersion. When flashers are tested as specified in 4.5.6, there shall be no damage that would affect subsequent operation.

3.10 Shock. After flashers are tested as specified in 4.5.7, they shall meet the requirements of 3.5.4.1, and 3.5.4.2, as applicable.

3.11 Altitude. When flashers are tested as specified in 4.5.8, there shall be no damage that would affect subsequent operation.

3.12 Humidity. After flashers are tested as specified in 4.5.9, they shall meet the requirements of 3.5.4.1 and 3.5.4.2, as applicable.

3.13 Salt spray. When flashers are tested as specified in 4.5.10, they shall meet the requirements of 3.5.4.1, 3.5.4.2, and 3.5.4.3, as applicable.

3.14 Sand and dust (when specified) (see 3.1). When flashers are tested as specified in 4.5.11, they shall meet the requirements of 3.5.4.1, 3.5.4.2, and 3.5.4.3, as applicable.

3.15 Electromagnetic interference. When flashers are tested as specified in 4.5.12, flashers shall meet the requirements of MIL-STD-461, equipment class IIIA. Following the test, flashers shall meet the requirements of 3.5.4.1 and 3.5.4.2, as applicable.

3.16 Transient voltage. When flashers are tested as specified in 4.5.13, there shall be no evidence of damage.

3.17 Life. When flashers are continuously operated as specified in 4.5.14 for the specified period of time (see 3.1), there shall be no failures. Following this test, flashers shall meet the requirements of 3.5.4.1, 3.5.4.2, 3.5.4.3, and 3.6, as applicable.

3.18 Marking. Flashers shall be marked in accordance with MIL-STD-1285, with the following information:

	EXAMPLE
MS or specification sheet part number	MS24577-1 or M26301/1-01
Item name	Flasher, Aircraft Navigational Light
Source code	12345
Functional diagram	See 3.4.7
Load rating	5.2 amp

3.19 Workmanship. Flashers shall be so fabricated as to be uniform in quality and shall be free from cracks, displaced parts, sharp edges, burrs, and other defects that will affect life, serviceability, and appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplier and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections.

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.2) on sample flashers produced with equipment and procedures normally used in production.

4.3.1 Sample size. Thirteen flashers shall be selected at random and subjected to qualification testing as specified in table II.

4.3.2 Inspection routine. Sample flashers shall be subjected to the inspections specified in table II, in the order shown. All flashers shall be subjected to group I inspection. Upon completion of group I inspection, the samples shall be subdivided as specified in table II for groups II, III, IV, and V.

4.3.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.3.4 Retention of qualification. To retain qualification, the supplier shall forward a report at 24 or 36 month intervals, as applicable (see 4.4.2.1.1), to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery (group A), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
(b) The results of tests performed for qualification verification inspection (group B) including the number and mode of failures. The report shall include results of all qualification verification inspection tests performed and completed during the 24- or 36-month period, as applicable. If the summary of test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 24- or 36-month period, as applicable, may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 24- or 36-month period, as applicable, that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the flasher. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative flasher of those covered by a single military standard or specification sheet, to testing in accordance with the qualification inspection requirements.

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TABLE II. Qualification inspection.

Examination or test	Requirement paragraph	Test paragraph	Number of sample flashers to be inspected	Number of failures
Group I				
Visual and mechanical examination (external)	3.1, 3.3, 3.4, 3.18 and 3.19	4.5.1	All relays in sample (13 total 1/)	1
Flash characteristics	3.5.4	4.5.2		
Voltage drop	3.6	4.5.3		
Group II				
High and low temperature soak and operation	3.7	4.5.4	3	
Vibration	3.8	4.5.5		
Immersion	3.9	4.5.6		
Group III				1
Shock	3.10	4.5.7	3	
Altitude	3.11	4.5.8		
Humidity	3.12	4.5.9		
Salt spray	3.13	4.5.10		
Sand and dust (when specified)	3.14	4.5.11		
Group IV				
Electromagnetic interference	3.15	4.5.12	2	0
Transient voltage	3.16	4.5.13		
Group V				
Life	3.17	4.5.14	4	0

1/ One additional flasher is included in the sample to allow for the one defective relay allowed for Group I.

4.4 Quality conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.4.1.1 Inspection lot. An inspection lot shall consist of all flashers covered by a single military standard or specification sheet, produced under essentially the same conditions within a period not to exceed 4 weeks, or 500 units, whichever occurs first, and offered for inspection at one time.

4.4.1.2 Group A inspection. Group A inspection shall consist of the examinations and test specified in table III in the order shown.

4.4.1.2.1 Sampling plan. Group A tests and examinations shall be performed on 100 percent of the flashers furnished under this specification.

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4.4.1.2.2 Rejected lots. The lot shall be rejected if more than 10 percent of the flashers fail to pass group A inspection. If an inspection lot is rejected, the supplier may rework it to correct the defects and resubmit for reinspection. Such lots shall be separated from new lots, and shall be clearly identified as reinspected lots. Individual flashers which are rejected from accepted lots may be reworked, when practicable, and may be resubmitted for reinspection.

4.4.1.2.3 Disposition of sample units. Sample flashers which have passed all the group A inspections may be delivered on the contract or purchase order, if the lot is accepted and the sample flashers are still within specified electrical tolerances.

TABLE III. Group A inspection.

Examination or test 1/	Requirement paragraph	Test paragraph
Flash characteristics - - - - -	3.5.4	4.5.2
Voltage drop - - - - - - - - - -	3.6	4.5.3
Visual and mechanical - - - - -	3.1, 3.3, 3.4, 3.18 and 3.19	4.5.1

1/ Flashers shall be tested at rated voltage and rated lamp load.

4.4.2 Qualification verification inspection. Qualification verification inspection shall consist of group B. Delivery of products which have passed group A inspection shall not be delayed pending the results of these qualification verification inspection.

4.4.2.1 Group B inspection. Group B inspection shall consist of the tests specified in table IV in the order shown. Group B inspection shall be made on sample units selected from inspection lots that have passed group A inspection.

TABLE IV. Group B inspection.

Test	Requirement paragraph	Test paragraph	Number of sample flashers to inspect	Number of failures
Group I				
High and low temperature soak and operation - - - - - - - - -	3.7	4.5.4	3	1
Vibration - - - - - - - - - - - -	3.8	4.5.5		
Immersion - - - - - - - - - - -	3.9	4.5.6		
Group II				
Shock - - - - - - - - - - - - - -	3.10	4.5.7	3	
Altitude - - - - - - - - - - - - -	3.11	4.5.8		
Humidity - - - - - - - - - - - -	3.12	4.5.9		
Salt spray - - - - - - - - - - -	3.13	4.5.10		
Sand and dust (when specified) -	3.14	4.5.11		
Group III				
Life - - - - - - - - - - - - - - -	3.17	4.5.14	4	0

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4.4.2.1.1 Sampling plan. Ten sample units shall be selected for each type of flasher produced, 24 months after the date of notification of qualification, and after each subsequent 36-month period. A suppliers normal quality control tests, production tests, environmental tests, and so forth, may be used to fulfill all or part of group B inspection; however, all of the group B inspection shall be completed as specified.

4.4.2.1.2 Failures. The number of allowable failures shall be as listed in table IV.

4.4.2.1.3 Disposition of sample flashers. Sample flashers which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.4.3 Noncompliance. If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample flashers (all inspection, or the inspection which the original flasher failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.4.4 Inspection of preparation for delivery. The sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Methods of examination and test. When applicable, testing of flashers with plug-in terminations shall be performed with the appropriate connector mated to the flasher.

4.5.1 Visual and mechanical examination.

4.5.1.1 External. Flashers shall be examined to verify that the materials, external design and construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.18 and 3.19).

4.5.2 Flash characteristics (3.5.4).

4.5.2.1 Flash rate (see 3.5.4.1). Flashers shall be tested at atmospheric pressure approximately 29.92 inches and at room temperature 23 ±5°C over the input voltage range specified in 3.5.2.

4.5.2.1.1 Single output circuits (see 3.5.4.1.1). Flashers shall be operated at rated voltage for 5 minutes for compliance with the flashing requirements specified (see 3.1) by loading in turn with the specified maximum rated and minimum rated lamp load. Flashers shall operate 80 percent of the time at maximum rated load and 20 percent of the time at minimum rated load. The flash rate shall be as specified (see 3.1).

4.5.2.1.2 Dual output circuits (see 3.5.4.1.2). Flashers shall be operated as specified in 4.5.2.1.1 except that the operation cycle shall be for 10 minutes. The flash rate shall be as specified (see 3.1).

4.5.2.2 Flash cycle (on to off ratio) (see 3.5.4.2). Flashers shall be tested at atmospheric pressure approximately 29.92 inches and at room temperature 23 ±5°C over the voltage range as specified in 3.5.2.

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4.5.2.2.1 Single output circuits (see 3.5.4.2.1). Flashers shall be operated for 5 minutes for compliance with the flash cycle requirements by loading in turn with a maximum and minimum lamp load as specified in table I. The flash cycle shall be as specified in 3.5.4.2.1.

4.5.2.2.2 Dual output circuits (see 3.5.4.2.2). Flashers shall be operated as specified in 4.5.2.2.1 except that the operation cycle shall be 10 minutes. The flash cycle shall be as specified in 3.5.4.2.2.

4.5.2.3 Switching time (see 3.5.4.3). Flashers shall be tested with rated input voltage and rated lamp load applied (see 3.1).

4.5.3 Voltage drop (see 3.6). Flashers shall be operated for 10 minutes each at minimum rated and maximum rated voltages, and maximum rated lamp load. Unless otherwise specified (see 3.1), the voltage drop in the lamp load shall not exceed 4.0 percent of the applied input voltage.

4.5.4 High and low temperature soak and operation (see 3.7). Environmental tests shall be conducted in accordance with the following procedures of MIL-STD-810.

4.5.4.1 High temperature (see 3.7.1). Flashers shall be subjected to high temperatures in accordance with procedure I, method 501, except that the soak period shall be for 24 hours in lieu of 48 hours. At the completion of the 24-hour soak period, the flashers shall be operated at the soak temperature (71°C) and tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.3, as applicable.

4.5.4.2 Low temperature (see 3.7.2). Flashers shall be subjected to low temperature in accordance with procedure I, method 502. Flashers shall be exposed to but not operated in, an ambient temperature of -62°C for a period of 48 hours. The temperature of the chamber shall be raised to -54°C and maintained for an additional 24-hour period or until temperature stabilization is reached. While the flashers are being maintained at -54°C, the minimum specified input voltage shall be applied to the input terminals and the minimum specified lamp load connected to the output. The flashers shall begin to operate at the specified flash rate (see 3.1) within 30 seconds after the input power is applied. Operation shall continue at the low temperature for a period of 10 minutes with the voltage held at the same rated value as was used to start the flasher. At the end of the test, flashers shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.3, as applicable, except that the flashers shall be operated at rated voltage and rated lamp load (see 3.1).

4.5.5 Vibration (see 3.8). Flashers shall be subjected to vibration in accordance with procedure I, curve AT, method 514, of MIL-STD-810, except the frequency range shall extend from 500 hertz to 1500 hertz at 10G's. During the entire test, the flashers shall be operated and connected to flash the lamp load specified in 4.5.14. Flashers shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.3, as applicable, except that flashers shall be tested at rated input voltage and rated lamp load (see 3.1).

4.5.6 Immersion (see 3.9).

4.5.6.1 Gasoline resistance. Flashers shall be immersed in a high-test gasoline of any grade conforming to MIL-G-5572 for a period of 1 hour at a temperature of 23 ±5°C. Flashers shall then be removed from the test fluid and allowed to drain for 1 hour at room temperature. There shall be no swelling of seals, gaskets or other components. There shall be no separation of components, dissolving of compounds or adhesives, nor deterioration that would cause the flashers to fail if exposed to gasoline in service.

4.5.6.2 Jet fuel resistance. Flashers shall be immersed in JP-4 fuel conforming to MIL-T-5624 for a period of 1 hour at a temperature of 23 ±5°C. Flashers shall then be removed and allowed to drain for 1 hour at room temperature. There shall be no swelling of seals, gaskets or other components. There shall be no separation of components, dissolving of compounds or adhesives, nor deterioration that would cause the flashers to fail if exposed to jet fuel in service.

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4.5.6.3 Hydraulic fluid resistance. Flashers shall be immersed in hydraulic fluid conforming to MIL-H-5606 for a period of 1 hour at a temperature of 23 ±5°C. Flashers shall then be removed from the test fluid and allowed to drain for 1 hour at room temperature. There shall be no swelling of seals, gaskets, or other components. There shall be no separation of components, dissolving of compounds or adhesives, nor deterioration that would cause the flasher to fail if exposed to hydraulic fluid in service.

4.5.7 Shock (see 3.10). Flashers shall be subjected to method 516, procedure I, of MIL-STD-810 (half-sine shock wave). Three shocks, each having an acceleration of 30 ±3G's and a duration of 9.0 ±2.0 milliseconds, shall be applied in each of the three mutually perpendicular directions. Flashers shall then be tested as specified in 4.5.2.1.1 and 4.5.2.1.2, as applicable, except that the flashers shall be operated at rated input voltage and rated lamp load (see 3.1).

4.5.8 Altitude (see 3.11). Flashers shall be subjected to altitude in accordance with procedure II, method 500, of MIL-STD-810. Reduce internal pressure to 1.32 in. of Hg or 70,000 feet above sea level.

4.5.9 Humidity (see 3.12). Flashers shall be subjected to humidity test in accordance with procedure I, method 507, of MIL-STD-810, except the temperature and humidity in the chamber shall be +49°C (+120°F) and 95 percent, respectively. At the end of the 240-hour period, the flashers shall be tested as specified in 4.5.2.1 and 4.5.2.2, as applicable, at rated input voltage and rated lamp load (see 3.1).

4.5.10 Salt spray (see 3.13). Flashers shall be subjected to the salt spray tests specified in procedure I, method 509, of MIL-STD-810. After exposure, the flashers shall be washed, shaken, air blasted, and then permitted to dry for 48 hours at room temperature. The flasher shall then be visually examined. There shall be no physical damage which would interfere with the operation or use of the flashers. Flashers shall then be tested as specified in 4.5.2.1.1 and 4.5.2.1.2, as applicable, except that flashers shall be operated at rated input voltage and rated lamp load (see 3.1). Flashers shall again be operated and inspected 48 hours later.

4.5.11 Sand and dust (when specified) (see 3.1 and 3.1.4). Flashers shall be tested as specified in method 510 of MIL-STD-810. Flashers shall then be tested as specified in 4.5.2.1.1 or 4.5.2.1.2, as applicable.

4.5.12 Electromagnetic interference (see 3.15). Flashers shall be subjected to electromagnetic interference as specified in MIL-STD-461, equipment class IIIA. During this test the flashers shall be operating an electrical load in accordance with 4.5.14. The radio noise caused by the flashers shall be within the specified limits both before and after the test. During tests, flashers shall be loaded in accordance with 4.5.14 except minimum rated lamp loads shall be applied. Following the tests, flashers shall meet the requirements of 4.5.2.1 and 4.5.2.2, as applicable, except flashers shall be tested at rated input voltage and rated lamp load (see 3.1).

4.5.13 Transient voltage (see 3.16). Flashers shall be subjected to a total of 80 transient voltage pulses applied to the input (control power) terminals. A resistive load of the applicable lamp voltage and current rating shall be connected to the output (lamp load) circuit. Flashers requiring dc control power shall be tested as follows: With the flasher operating at normal voltage, 20 voltage pulses shall be applied across the input terminals when the flasher output circuit is in the nonconducting or "Load OFF" condition. Negative polarity shall be applied to the flasher "Positive" terminal, and positive polarity to the flasher "Negative" terminal. Repeat test with 20 additional pulses applied when the flasher is in conducting or "Load ON" condition. Pulse voltage shall be $286\ ^{+0}_{-5}$ percent of rated voltage. Forty additional pulses shall be applied as above except that a positive pulse voltage shall be applied to the flasher control "Positive" terminal and a negative pulse voltage to the "Negative" terminal. Duration of applied transient voltage shall be 0.1 second, minimum. Flashers requiring ac control power shall be tested similarly to the flasher units having dc control power except that only 40 pulses are required. The peak pulse voltage shall be $166\ ^{+0}_{-5}$ percent of the rated rms voltage. Twenty pulses shall be applied when the flasher output is in the nonconducting of "Load OFF" condition and 20 pulses shall be applied when the flasher output circuit is in the "Load ON" condition. Duration of the ac pulse transient shall be 0.30 second minimum. After the specified pulses have been applied, the flashers shall be tested to determine compliance with 4.5.2.1 and 4.5.2.2, as applicable.

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4.5.14 Life (see 3.17). Flashers shall be subjected to a continuous operational life test. The load shall consist of aircraft lamps of adequate number and connected in parallel to provide a steady state current 100 ±5.0 percent of maximum rated lamp load. Input voltage shall be rated source voltage. The first 30 percent of the life test shall be conducted at the maximum rated ambient temperature and the last 70 percent of the life test shall be performed at ambient room temperature. The test shall be conducted for the specified hours (see 3.1). At the completion of the life test, flashers shall be tested to determine compliance with 4.5.2.1, 4.5.2.2, 4.5.2.3, and 4.5.3, as applicable, except rated input voltage and rated lamp load shall be applied (see 3.1).

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Flashers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Flashers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Flashers shall be individually packaged in accordance with the methods of MIL-P-116 specified herein insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein.

5.1.1.4.1 Hermetically sealed flashers. Hermetically sealed flashers shall be packaged in accordance with method III. The unit container shall conform to PPP-B-566 or PPP-B-676.

5.1.1.4.2 Nonhermetically sealed flashers. Unless otherwise specified (see 6.2), nonhermetically sealed flashers shall be packaged in accordance with submethod IA-15. The container shall conform to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Flashers shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The packaged flashers shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged flashers shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged flashers shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

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5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Flashers, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Flashers, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Flashers, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.1), each unit package, exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General. Special requirements for Army and Navy procurements are specified in 5.4.2 and 5.4.3, respectively.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistant with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Army procurements.

5.4.2.1 Level A unit packaging. MIL-P-116 submethod IC-2 shall be used in lieu of submethod IA-15. All unit containers shall either be weather (or water) resistant or overwrapped with waterproof barrier materials. Containers conforming to PPP-B-566 or PPP-B-676 shall be overwrapped with waterproof barrier materials or shall conform to MIL-B-43014 (see 5.1.1.4.1, 5.1.1.4.2, and 6.1).

5.4.2.2 Level A and B packing. For level A packing when quantities per destination are less than a unitized load, the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4, or PPP-B-585, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish B. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

5.4.2.3 Level A and B unitization. For level A and B unitization, the fiberboard caps shall be weather resistant and softwood pallets conforming to NN-P-71, type IV, size 2, shall be used (see 5.2.4.1 and 5.2.4.2).

5.4.3 Navy requirements. For Navy requirements the use of polystyrene loose fill material (such as strips, strands, and beads) is prohibited for packaging and packing applications.

5.4.4 Air Force requirements. For Air Force requirements submethod IC-2 shall be used in lieu of submethod IA-15 (see 5.1.1.4 and 6.1).

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6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable MS standard or specification sheet and the complete MS part number or specification sheet part number.
(c) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(d) Method of preservation, if other than submethod IA-15 (see 5.1.1.4.2, 5.4.2.1 and 5.4.4).
(e) Special marking, if required (see 5.3).

6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activities responsible for the Qualified Products List is the Air Force Logistics Command, Code 80, Detachment 21, SMAMA/MMREE, AFG59 Engineering and Technical Support Branch, Directorate of Material Management, Wright-Patterson AFB, OH 45433, except MS25205(AS) for which the Naval Air Systems Command, AIR 53353, Washington, D.C. 20360, will be responsible.

6.3 Definitions.

6.3.1 Control power. Control power is considered the power which is used to produce the required cycling rate and duration of power pulses to the lamp load.

6.3.2 Lamp power. Lamp power is the power which is switched on and off within the flasher and which energizes the load (lamp).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

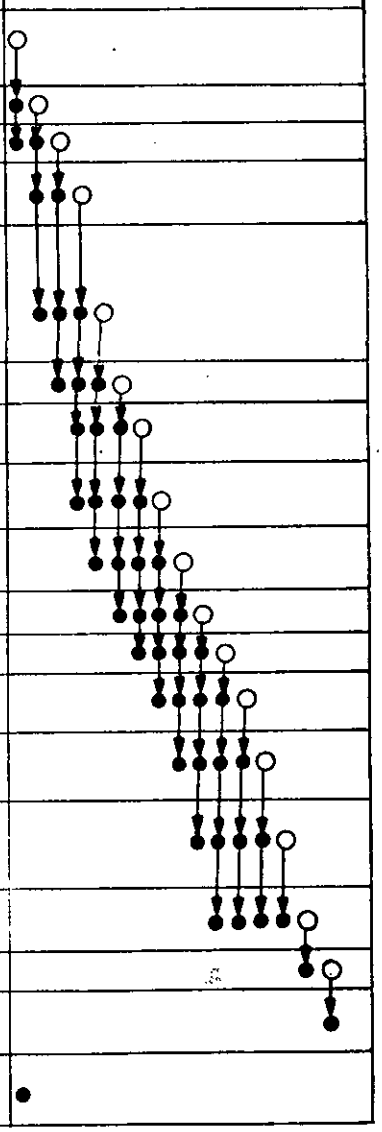
6.5 Intermetallic contact. The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, since intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table V. Table V shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient temperatures. All members of a group are considered as completely compatible, one with the other. Compatible couples between groups have been specified in table V based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table V shows, in addition to EMF against a calomel electrode, a derived "anodic index" with Group 1 (gold, etc.) as 0 and Group 18 (magnesium, etc.) as 175. Subtraction of a lower group anodic index gives the EMF difference in hundredths of a volt.

6.5.1 Groups. Table V sets up 18 primary groups. It may be noted that neither the metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples. All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action, when coupled with any member within the group; for example, such dissimilar metals as platinum and gold. Similarly, such basically dissimilar alloys as austenitic stainless steel, silver-solder, and low brass (all members of Group 5) are inherently nonsusceptible when coupled together.

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TABLE V. Compatible couples (see 6.5) 1/

Group No.	Metallurgical category	EMF (volt)	Anodic index (0.01 v)	Compatible couples
1	Gold, solid and plated; gold-platinum alloys; wrought platinum (most cathodic)	+0.15	0	
2	Rhodium plated on silver-plated copper	+0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	-0.15	30	
5	Copper, solid or plated; low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	-0.20	35	
6	Commercial yellow brasses and bronzes	-0.25	40	
7	High brasses and bronzes; naval brass; Muntz metal	-0.30	45	
8	18 percent chromium type corrosion-resistant steels	-0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	-0.45	60	
10	Tin-plate; terneplate; tin-lead solder	-0.50	65	
11	Lead, solid or plated; high lead alloys	-0.55	70	
12	Aluminum, wrought alloys of the duralumin type	-0.60	75	
13	Iron, wrought, gray, or malleable; plain carbon and low alloy steels, armco iron	-0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	-0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	-0.80	95	
16	Hot-dip-zinc plate; galvanized steel	-1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	-1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	-1.60	175	



1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

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6.5.2 Compatibility graphs. Permissible couple series are shown in table V by the graphs at the right. Members of groups connected by lines will form permissible couples. A O indicates the most cathode member of each series, a ● an anodic member, and the arrow indicates the anodic direction.

6.5.3 Selection of compatible couples. Proper selection of metals in the design of equipment will result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt, which is not allowable by table V. In this case, other metals or plates will be required. It should be noted that, in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium. If the surface area of the cathodic part is significantly greater than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly intensified. Material selection for intermetallic contact parts, therefore, should establish the smaller part as the cathodic member of the couple, whenever practicable.

6.5.4 Plating. When base metals intended for intermetallic contact form couples not allowed by table V, they are to be plated with those metals which will reduce the potential difference to that allowed by table V.

Custodians:
Air Force - 80
Army - AV
Navy - AS

Review activities:
Air Force - 11
Army - EL
Navy -
DSA - ES

User activities:
Air Force -
Army -
Navy - MC

Preparing activity:
Air Force - 80

Agent:
DSA - ES

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